Shashank Jatav

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AI Leader/Innovator/Data Scientist

AI/ML leader with 13+ years of experience building production-grade machine learning, NLP, and GenAI systems across healthcare, life sciences, legal tech, and automotive domains. Proven record of taking AI products from 0→1 and scaling them in production—including LLM fine-tuning, RAG over millions of documents, model observability, scientific AI, and revenue-linked data products. Led teams from 4 to 20, partnered with cross-functional stakeholders and scientists at Bosch, Yale, University of Toronto, and Pfizer, and contributed to 6+ publications, 2 patents, and 10 abstracts. Pioneer of explainable and reliable AI systems in production.

Core Expertise

GenAI & LLM Products • Retrieval-Augmented Generation (RAG) • Fine-Tuning & Evaluation • NLP & Information Extraction • Medical Coding AI • AI Strategy & Team Leadership • MLOps, Monitoring & Observability • Knowledge Graphs • Biomedical / Healthcare AI • Data Products & Revenue Ownership • Python, PyTorch, TensorFlow, Scikit-learn, SQL, AWS, GCP, Kubernetes

WORK EXPERIENCE

Arintra, Bengaluru Designation: Principal ML Engineer *Sep 24 – Present*

Job Description: **Principal for all NLP initiatives/Agentic AI and medical coding product**

- **Led the development** of medical coding product of new specialities like radiology, primary care etc. Initiated using fine tuned models for medical coding increasing the accuracy from 40 to 90 for EnM coding
- **Optimized the LLM costs** by more than 70% by switching to fine-tuned models and local models such as qwen, kimi-k2
- **Implemented RAG solution** for all specialities to generate ICD candidates for prompt leading to higher accuracy of 90% for ICDs
- **Conceptualized and led** the implementation of LLM observability to track any production errors, model and data drifts
- **Led the development** of tooling for data annotation team both internal and external enhancing their productivity
- **Developed an automated bug detector** reducing manual work done for detecting bugs in pipeline by 70%
- **Conceptualized and developing** an automation model and model confidence framework linking accuracy improvements in models to actual automation in the product
- **Developed LLMOps platform** reducing the development time multifold by 400%. Platform included self-serve systems, automated prompt generation algorithms and training data generation systems.

Sirion, Gurugram Designation: Technical Lead II *Mar 24 – Sep 24*

Job Description: **Technical lead for GenAI products**

- **Ingested millions of documents** into **RAG** architecture within 3 days as compared to a month timeline
- **Improved the latency of hybrid RAG by 300%** leading to product adoption and reduced failure rate by 100%, the RAG covered millions of documents
- **Developed multiple no-code GenAI POCs** using langflow, Gradio, knowledge graph for domain users such as lawyers
- **Contributed** to development of Sirion's marquee product - contract redlining

Arintra, Bengaluru Designation: Director NLP *Aug 23 – Mar 24*

Job Description: **Leading NLP team and defining tech strategy**

- **Defined the NLP strategy** of the product leading to increase in accuracy from 70 to **more than 95**
- **Led the team to be highly performant** and shipping LLMs and agents into production**,** implemented monitoring and alerting to reduce **variance in LLM output by 100%**
  - **Led the implementation using RAG** covering millions of documents and concepts leading to low hallucination
- **Fine-tuned open source LLMs** to create an alternative to commercial APIs to **reduce the production costs by 200%**

Elucidata, New Delhi Designation: Advisor AI/ML *May 23 – Jul 23*

Job Description: **Advising on LLM Strategy and team**

- **Advised** on using LLMs for data curation leading to an increase in F1 score beyond 90 and surpassing human accuracy

Elucidata, New Delhi Designation: Director, Data Products *Nov 20 – Apr 23*

Job Description: **Leading all data product development**

- **Leading** ingestion, curation, quality control of over **a million datasets** on the platform
- **Formulated** ML data offering leading to product revenue growth from **100k to 1 million dollars ARR**
- **Grew** the team from **4 to 20**, created a ML team from scratch doing **deep learning NLP (SLMs) and MLOps**
- **Championed** the idea of data centric NLP for curation leading to reduction of curation costs by **100%**

Elucidata, New Delhi Designation: Senior Data Scientist *Nov 18 – Nov 20*

Job Description: **Leading innovation projects and key product initiatives**

- **Initiated** multiple ML projects in the organisation that included **deep learning NLP (SLMs) (BERT based) and Bioinformatics ML** leading to first **100k dollars revenue** from ML
    - **Led the development** of a random forest model for identifying bad peaks from good ones in mass spectrometry data leading to reduction in time for curation of peaks by 200%
- **Streamlined** data science practices in organisation using modern development tools leading to **TAT reduction by 100%**
- **Conceptualised and led** the product development of first data offering for the organisation resulting in organization level pivot and ML based IP creation that led to **funding round of 5 million dollars and Series A of 16 million dollars**

Elucidata, New Delhi Designation: Senior Data Scientist *Nov 16 – Nov 18*

Job Description: **Responsible for initial service projects, creating a parallel revenue stream**

- **Created** service led revenue stream from **zero to 500k dollars**, work involved analyzing public and proprietary datasets
- Found a **therapeutic target using ML** for an early-stage startup leading to a **million dollar revenue**
    - **Developed an SVM model** classifying sample into blood cell states leading to identification of target
- **Collaborated** with scientists from top universities and Pharma such as **Yale, Toronto, Pfizer, UFL** leading to multiple publications

Bosch Limited, Bangalore Designation: Assistant Manager (Engineering Powertrain R&D) *July 13 – Nov 16*

Job Description: **Responsible for engine calibration and strategy development for Indian future emission norms**

- **Filed 2 Indian Patents** for increasing volumetric efficiency of engines, **authored 6 invention reports, 2 development reports**
- Developed strategy is being **incorporated on 2,00,000 vehicles** all over India for stringent BS4 emission norms
- **Defined and assessed** a novel air system for meeting stringiest emission norm in the world for single cylinder NA diesel engines

EDUCATION

Degree/Certificate	Institution	Year	CPI/%
B. Tech., Mechanical Engineering	Indian Institute of Technology, Kanpur	2013	8.3
XII: ISC	City Montessori School, Lucknow	2009	92.75

An average grade of **9.0** in all the departmental courses

Publications (See more here)

- Integrated Metabolomic and Transcriptomic Profiling Reveals Novel Activation-Induced Metabolic Networks in Human T cells bioRxiv 635789; doi: https://doi.org/10.1101/635789
- Metabolic regulation by PD-1 signaling promotes long-lived quiescent CD8 T cell memory in mice. Sci Transl Med. 2021. doi: https://doi.org/10.1126/scitranslmed.aba6006. Epub 2021 Oct 13. PMID: 34644150; PMCID: PMC8896520.
- Reducing Inference Time of Biomedical NER Tasks using Multi-Task Learning (Accepted in ICON 2022, indexed in ACL)

Patents

- A method to improve the volumetric efficiency of single cylinder internal combustion engine, application no: 20164102518
- A method to improve the volumetric efficiency of single cylinder internal combustion engine, application no: 7135/CHE/2015

Presentations

- Watch me talk about team that I have built here
- My talk at Elucidata's User Group Meeting about our IP and curation
- See the talk about data centric NLP here, represented Elucidata at conferences such as KGC, ASMS, AMIA and several others
- Participated in drug discovery hackathon organized by Gov of India – selected in top 5 teams out of 400 for moonshot approach
- Judged as one of the top 6 teams out of 2500 teams in Tata Motors Challenge Mindrover 2013

Courses/Certifications

Deep learning specialization (Coursera) – Certificate | Machine Learning (Coursera) – Certificate | Functional Programming Principles in Scala – Certificate | End-to-End Machine Learning with Tensorflow on Google Cloud (Coursera) – Certificate | Strategic Leadership Program for Business Leaders (SP Jain School of Global Management) | Apache Spark with Scala (Udemy) – Certificate | Digital Product Management (Coursera) – Certificate

Miscellaneous

- Awarded National Talent Search Scholarship in 2007 among 0.1 million students
- Awarded Self Starter title in Elucidata as recognition for initiatives in 2018
- Awarded Mad Scientist title in Elucidata as recognition for disrupting innovations in 2021